

# Determinants

## \* Introduction to Determinants

→ The determinant is a single scalar value associated with a square Matrix.

→ Determinant of a matrix A is denoted by  $\det(A)$  or  $|A|$

→ If  $\det(A) = 0 \Leftrightarrow$  Matrix is Singular

If  $\det(A) \neq 0 \Leftrightarrow$  Matrix is Non-Singular

→ Value of determinant changes when we apply elementary row operations on a Matrix.

$$\det(A) \neq \det(\text{REF}(A)) \quad \leftarrow \text{Why do we reduce A to REF(A)?}$$

→ determinant tells the scaling factor of area/volume of space and Orientation when a transformation A is applied.

## \* Computing Determinant of a Matrix

→ There are 2 ways to solve a Determinant

### ① Using Cofactor

→ The  $\det(A)$  is a combination of any row  $i$  times its cofactor

$$\det(A) = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}$$

and  $C_{ij} = (-1)^{i+j} \det(M_{ij})$

### ② Using Row reduction.

→ Convert matrix to row echelon form (upper triangle)

→ Determinant = Product of diagonal elements

→ Adjust Sign if rows are swapped. why?

### Effect of Row Operations on Determinant

| Row Operation                      | Effect on Determinant |
|------------------------------------|-----------------------|
| Swap two rows                      | Multiplies by -1      |
| Multiply a row by k                | Multiplies by k       |
| Add multiple of one row to another | No change             |

Example: Compute  $\det(A)$  where  $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 6 & 10 \end{bmatrix}$

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

→ ① using cofactor

$$\begin{aligned} \det(A) &= \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 6 & 10 \end{vmatrix} = 1 \times \begin{vmatrix} 5 & 8 \\ 6 & 10 \end{vmatrix} + 2 \times (-1) \begin{vmatrix} 2 & 8 \\ 3 & 10 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} \\ &= 1 \times 2 - 2 \times (-4) + 3 \times (-3) \\ &= 2 + 8 - 9 = 1 \end{aligned}$$

② using row reduction

$$\det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 6 & 10 \end{vmatrix} \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix} = 1 \times 1 \times 1 = 1$$

Example: Compute  $\det(A)$  where  $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$

→ ① using cofactor

$$\det(A) = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 2(+5) + 3(-4) = 10 - 12 = -2$$

↑  
along row 1

$$\det(A) = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 4(-3) + 5(+2) = -12 + 10 = -2$$

↑  
along row 2

② using row reduction

$$\det(A) = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} \xrightarrow{R_2 = R_2 - 2R_1} \begin{vmatrix} 2 & 3 \\ 0 & -1 \end{vmatrix} = 2 \times (-1) = -2$$

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### \* Some interesting Property of Determinant

- ① The determinant of Identity matrix is 1
- ② The determinant changes sign when 2 rows are exchanged.
- ③ The determinant depends linearly on one row at a time.

$$\text{additivity} \rightarrow \begin{vmatrix} a & b & c \\ d & e & f \\ g+x & h+y & i+z \end{vmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} + \begin{vmatrix} a & b & c \\ d & e & f \\ x & y & z \end{vmatrix}$$

$$\text{Homogeneity} \rightarrow \begin{vmatrix} a & b \\ tc & td \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

- ④ If 2 rows of A are equal, then  $\det(A) = 0$

⑤ Subtracting a multiple of one row from another row leaves the same determinant

⑥ If  $A$  has a row of zeros, then  $\det(A) = 0$

⑦ Determinant of a Triangular Matrix is product of its diagonal elements

⑧  $\det(AB) = \det(A) \cdot \det(B)$

⑨  $\det(A^T) = \det(A)$

## What does $\det(A)$ tell us?

( $A$  is an  $n \times n$  matrix)

| Nonsingular ( $\det(A) \neq 0$ )            | Singular ( $\det(A) = 0$ )                   |
|---------------------------------------------|----------------------------------------------|
| $A$ is invertible.                          | $A$ is not invertible.                       |
| The columns are independent.                | The columns are dependent.                   |
| The rows are independent.                   | The rows are dependent.                      |
| The determinant is not zero.                | The determinant is zero.                     |
| $Ax = 0$ has one solution $x = 0$ .         | $Ax = 0$ has infinitely many solutions.      |
| $Ax = b$ has one solution $x = A^{-1}b$ .   | $Ax = b$ has no solution or infinitely many. |
| $A$ has $n$ (nonzero) pivots.               | $A$ has $r < n$ pivots.                      |
| $A$ has full rank $r = n$ .                 | $A$ has rank $r < n$ .                       |
| The reduced row echelon form is $R = I$ .   | $R$ has at least one zero row.               |
| The column space is all of $\mathbb{R}^n$ . | The column space has dimension $r < n$ .     |
| The row space is all of $\mathbb{R}^n$ .    | The row space has dimension $r < n$ .        |
| All eigenvalues are nonzero.                | Zero is an eigenvalue of $A$ .               |

## \* Geometric Meaning \*\*

→  $A_{m \times n} x = b$  ← A matrix  $A \in \mathbb{R}^{m \times n}$  is multiplied by a column vector  $x \in \mathbb{R}^n$ , results in a column vector  $b \in \mathbb{R}^m$ .

Matrix A transforms entire  $\mathbb{R}^n$  space and therefore every vectors in  $\mathbb{R}^n$

→ A represents a Linear Transformation

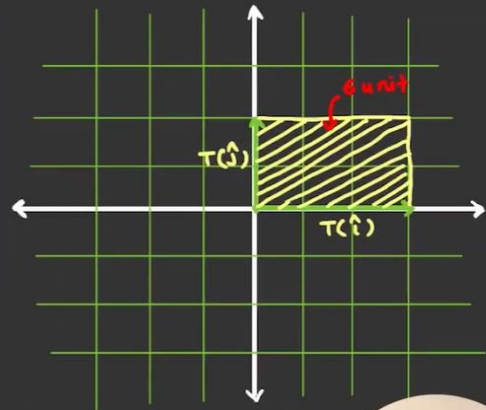
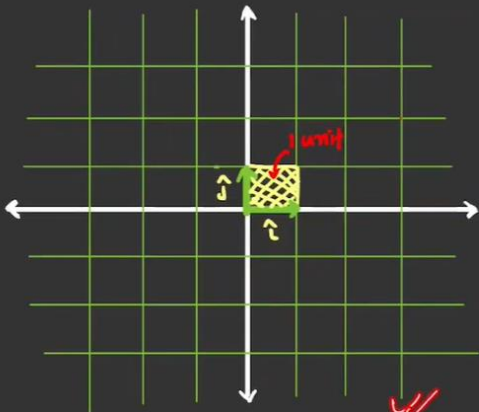
→  $\det(A)$  represents the signed scaling factor of area/volume produced by a linear transformation.

→ In 2D → area scaling  
In 3D → volume scaling

Here  $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ . why?

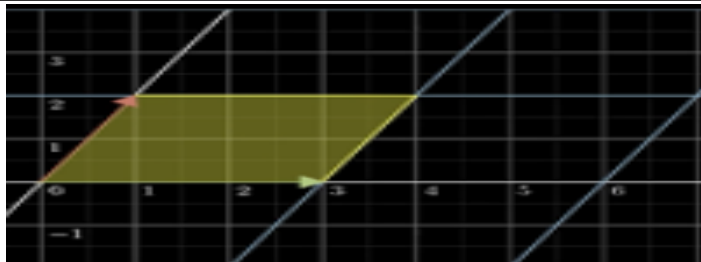
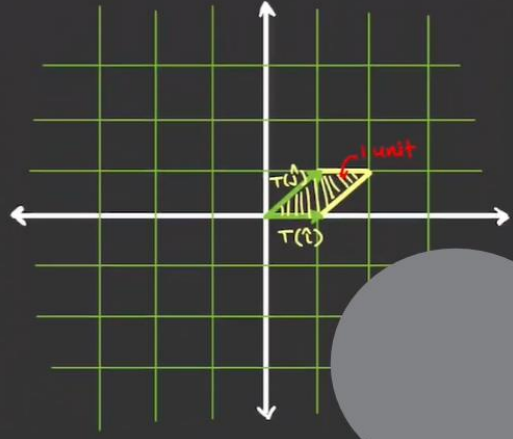
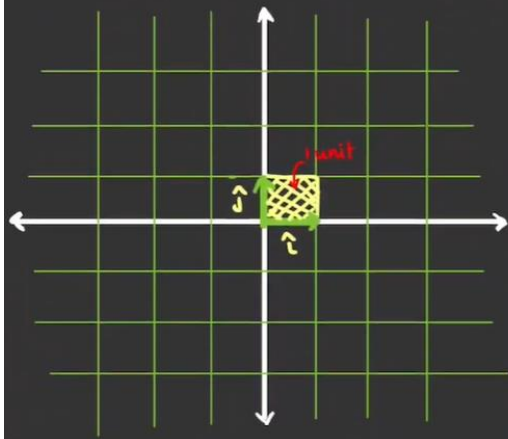
Example: Visualize the Linear Transformation when  $A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$

→  $\det(A) = \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix} = 6$  ← Linear transformation scales area by 6 times and +ve magnitude means orientation is preserved.



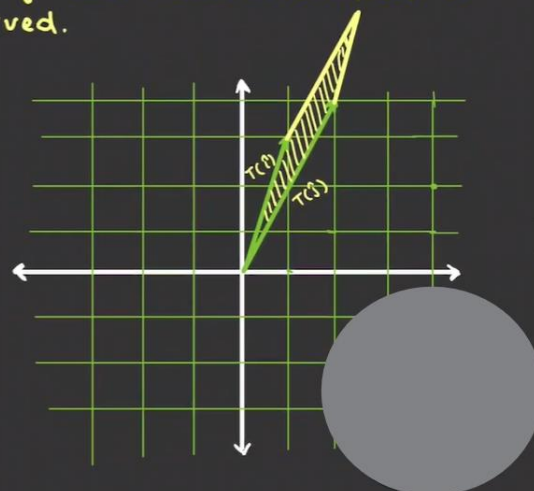
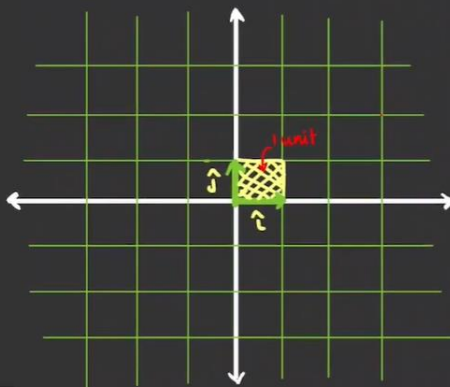
Example: Visualize the Linear Transformation when  $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\rightarrow \det(A) = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1$   $\leftarrow$  Linear transformation does not scale area and +ve magnitude means orientation is preserved.



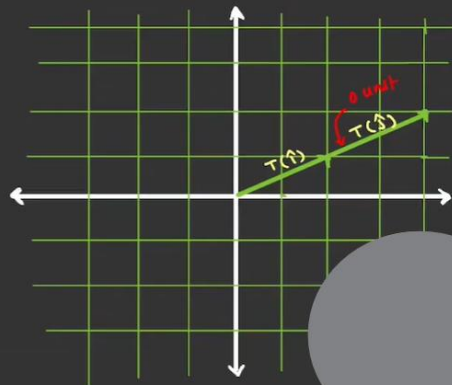
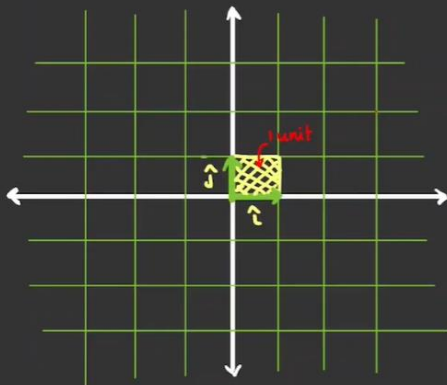
Example: Visualize the Linear Transformation when  $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$\rightarrow \det(A) = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = -2$   $\leftarrow$  Linear transformation scales area by 2 times and -ve magnitude means orientation is not preserved.



Example: Visualize the Linear Transformation when  $A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$

→  $\det(A) = \begin{vmatrix} 4 & 2 \\ 2 & 1 \end{vmatrix} = 0 \leftarrow$  Linear transformation brings Area=0



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Sign of determinant

$\det(A) > 0 \Rightarrow$  orientation preserved

$\det(A) < 0 \Rightarrow$  orientation reversed

$\det(A) = 0 \Rightarrow$  Area/volume becomes 0

↑ in  $\mathbb{R}^2 \rightarrow$  square become line  
in  $\mathbb{R}^3 \rightarrow$  cube become plane

### \* Application of Determinants \*

#### ① Computation of Inverse of a matrix

→ For a matrix A

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \frac{1}{|A|} \text{Cof}(A)^T$$

→ Matrix A is invertible if and only if A is non-singular ( $|A| \neq 0$ )

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$M_{ij}$  is the determinant of the submatrix obtained by deleting the i-th row and j-th column.

↩ ↪

Example: Calculate the inverse of  $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

→ Let's compute the cofactor matrix  $C$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\text{Adj}(A) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{and } |A| = 1$$

$$A^{-1} = \frac{1}{|A|} \text{adj}(A) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

## ② The Solution of $Ax=b$

→ For a system of Linear equations with

①  $n$  equations and  $n$  unknowns

② having unique solution

we can find the solution using Cramer's rule

→ The  $j$ th component of  $x = A^{-1}b$  is the ratio

$$x_j = \frac{\det(B_j)}{\det(A)}$$

where  $B_j = \begin{bmatrix} a_{11} & a_{12} & \dots & b_1 & \dots & a_{1n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & b_n & \dots & a_{nn} \end{bmatrix}$  has  $b$  in column  $j$   
 ↑  
 modified version  
 of  $A$

Example: Solve the systems of Linear equations using Cramer's Rule.

$$x_1 + 3x_2 = 0$$

$$2x_1 + 4x_2 = 6$$

→ Matrix form  $Ax=b$   $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 6 \end{bmatrix}$

$$x_1 = \frac{\det(B_1)}{\det(A)} = \frac{-18}{-2} = 9$$

$$x_2 = \frac{\det(B_2)}{\det(A)} = \frac{6}{-2} = -3$$

$$|A| = \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = 4 - 6 = -2$$

$$|B_1| = \begin{vmatrix} 0 & 3 \\ 6 & 4 \end{vmatrix} = 0 - 18 = -18$$

$$|B_2| = \begin{vmatrix} 1 & 0 \\ 2 & 6 \end{vmatrix} = 6 - 0 = 6$$

Hence  $x = \begin{bmatrix} 9 \\ -3 \end{bmatrix}$